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Learning Objectives of This Lecture

1. Proficiently formulate the **state equations** and **output equations** for continuous and discrete-time systems, and draw **signal flow graphs**;
2. Understand the definitions of variables (e.g., **characteristic matrix**, **transition matrix**, **system function**) in state variable analysis;
3. Understand the **s-domain** and **time-domain** solution methods for the state equations and output equations of continuous-time systems;
4. Understand the definitions of the **characteristic matrix** and **transition matrix** for discrete-time systems, as well as the **time-domain/z-domain** solution methods for their state equations and output equations.

8.1 Introduction

- The analysis methods for signals and systems discussed in previous chapters are the **input-output description method** (or **external method**), describing system characteristics through the relationship between input and output variables. Once a **differential/difference equation** is established for the system, it no longer concerns the system's internal conditions - **it only considers how the system's time/frequency characteristics (system function) influence the output**. This method is suitable for simple systems (e.g., single-input single-output) and cannot fully reveal internal system properties.
- To analyze complex systems (e.g., **nonlinear, time-varying, multi-input multi-output** systems), the **state variable description method** (or **internal method**) is used. It describes the system's internal variables via state variables, and connects input/output variables through these state variables to characterize the system's external behavior.
- **State variable**: The **minimal** set of variables that describe the system's state. It essentially reflects changes in the system's internal energy storage state.
- **state**: If the system's state variables at $t = t_0$ and the input for $t \geq t_0$ are known, the system's state and output at any $t \geq t_0$ can be fully determined. **An n th-order system has n initial states.**

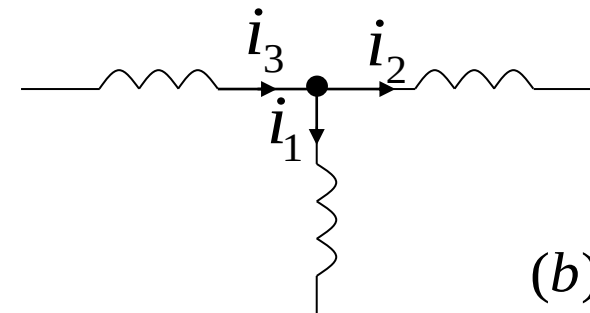
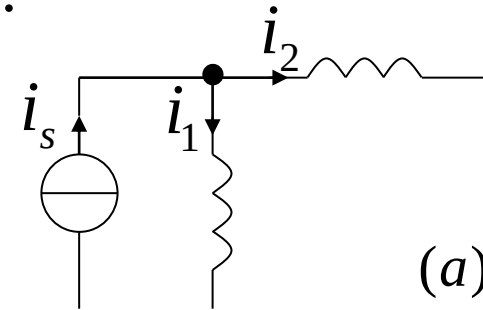
In circuits (a) and (b) shown (with the current source's current known), there are () independent inductor currents (serving as the system's state variables) in each circuit respectively.

A 2 and 3

B 2 and 2

C 1 and 2

D 1 and 1

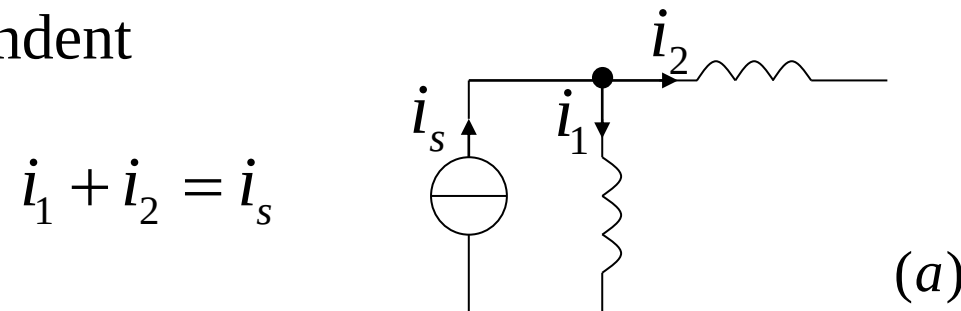


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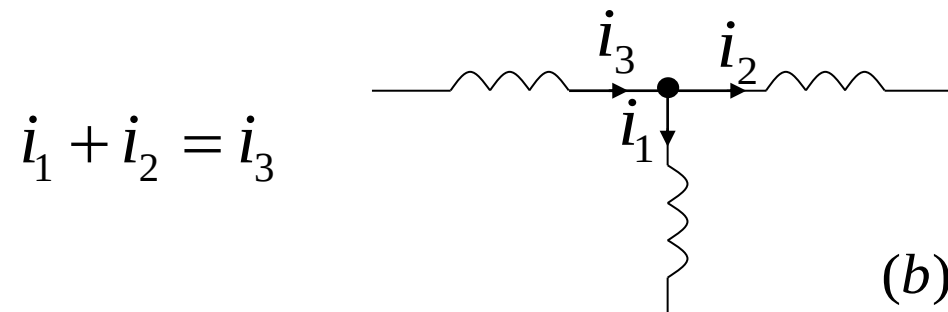
8.1 Introduction

State variables are mutually **independent** — e.g., an "**independent**" inductor current **cannot be fully expressed** by other inductor currents.

In Figure (a): Due to the constraint of the current source, only **1** of the two inductor currents is independent



In Figure (b): **2** of the three inductor currents are independent



Similarly, a "**independent**" capacitor voltage **cannot be fully expressed** by other capacitor voltages.

Main Advantages of State Variable Analysis:

- 1) It facilitates studying the variation laws of **physical quantities inside the system** - these quantities can be represented by a component of the state vector.
- 2) It applies to LTI single-input single-output systems, and also to describing the characteristics of **nonlinear, time-varying, multi-input multi-output systems**.
- 3) It is independent of the system's complexity: the mathematical models of complex and simple systems are similar, both expressed as linear combinations of state variables.
- 4) **State equations are first-order differential or difference equations**, which are convenient for **computer** analysis and calculation.
- 5) The main parameters of state equations clearly characterize the system's **key performance**, and can be used to analyze the system's controllability, observability, and stability.

8.2.1 Formulating State Equations Directly from a Circuit

- 1) Select state variables: Usually, **independent** inductor currents and **independent** capacitor voltages in the circuit are chosen as state variables.
- 2) Formulate equations: Write loop voltage equations for branches containing independent inductors, and node current equations for branches containing independent capacitors.
- 3) Arrange equations: Rearrange the obtained equations into the standard form.

8.2 Formulation of State Equations for Continuous-Time Systems

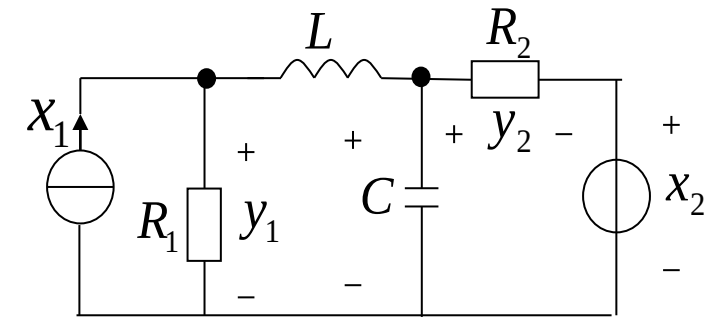


Example 8-1: For the circuit shown, take the current of the current source and the voltage of the voltage source as inputs, and take the voltages across the two resistors as outputs. Formulate the circuit's state equations and output equations.

Solution: This is a **2×2** system.

1) Choose the **inductor current** and **capacitor voltage** as **state variables**:

$$\lambda_1(t) = i_L(t) \quad \lambda_2(t) = v_c(t)$$



2) Formulate **state equations**:

Write the loop voltage equation including the current source, inductor, and capacitor:

$$L \frac{di_L(t)}{dt} = [x_1(t) - i_L(t)]R_1 - v_c(t)$$

Write the node current equation for the branch containing the capacitor,

$$C \frac{dv_c(t)}{dt} = [x_2(t) - v_c(t)] / R_2 + i_L(t)$$

3) Arrange the Equations

$$L \frac{di_L(t)}{dt} = [x_1(t) - i_L(t)]R_1 - v_c(t) = -R_1 i_L(t) - v_c(t) + R_1 x_1(t)$$

$$\therefore \frac{di_L(t)}{dt} = -\frac{R_1}{L} i_L(t) - \frac{1}{L} v_c(t) + \frac{R_1}{L} x_1(t)$$

$$C \frac{dv_c(t)}{dt} = [x_2(t) - v_c(t)] / R_2 + i_L(t) = i_L(t) - \frac{1}{R_2} v_c(t) + \frac{1}{R_2} x_2(t)$$

$$\therefore \frac{dv_c(t)}{dt} = \frac{1}{C} i_L(t) - \frac{1}{R_2 C} v_c(t) + \frac{1}{R_2 C} x_2(t)$$

8.2 Formulation of State Equations for Continuous-Time Systems



Write in standard form:

$$\frac{d\lambda_1(t)}{dt} = -\frac{R_1}{L}\lambda_1(t) - \frac{1}{L}\lambda_2(t) + \frac{R_1}{L}x_1(t)$$

$$\frac{d\lambda_2(t)}{dt} = \frac{1}{C}\lambda_1(t) - \frac{1}{R_2C}\lambda_2(t) + \frac{1}{R_2C}x_2(t)$$

Write in matrix form:

$$\begin{pmatrix} \frac{d\lambda_1(t)}{dt} \\ \frac{d\lambda_2(t)}{dt} \end{pmatrix} = \begin{pmatrix} -\frac{R_1}{L} & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{R_2C} \end{pmatrix} \begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \end{pmatrix} + \begin{pmatrix} \frac{R_1}{L} & 0 \\ 0 & \frac{1}{R_2C} \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$

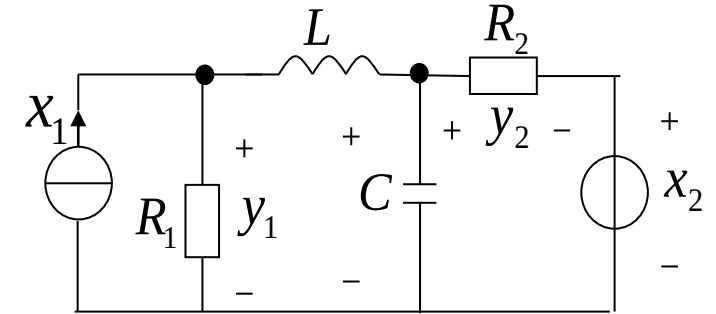
8.2 Formulation of State Equations for Continuous-Time Systems



4) Formulate the output equations

$$y_1(t) = [x_1(t) - i_L(t)]R_1$$

$$y_2(t) = v_c(t) - x_2(t)$$



then

$$y_1(t) = -R_1 i_L(t) + R_1 x_1(t) = -R_1 \lambda_1(t) + R_1 x_1(t)$$

$$y_2(t) = \lambda_2(t) - x_2(t)$$

Write in matrix form

$$\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix} = \begin{pmatrix} -R_1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \end{pmatrix} + \begin{pmatrix} R_1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$

8.2 Formulation of State Equations for Continuous-Time Systems

8.2.2 Formulating State Equations from Input-Output Description

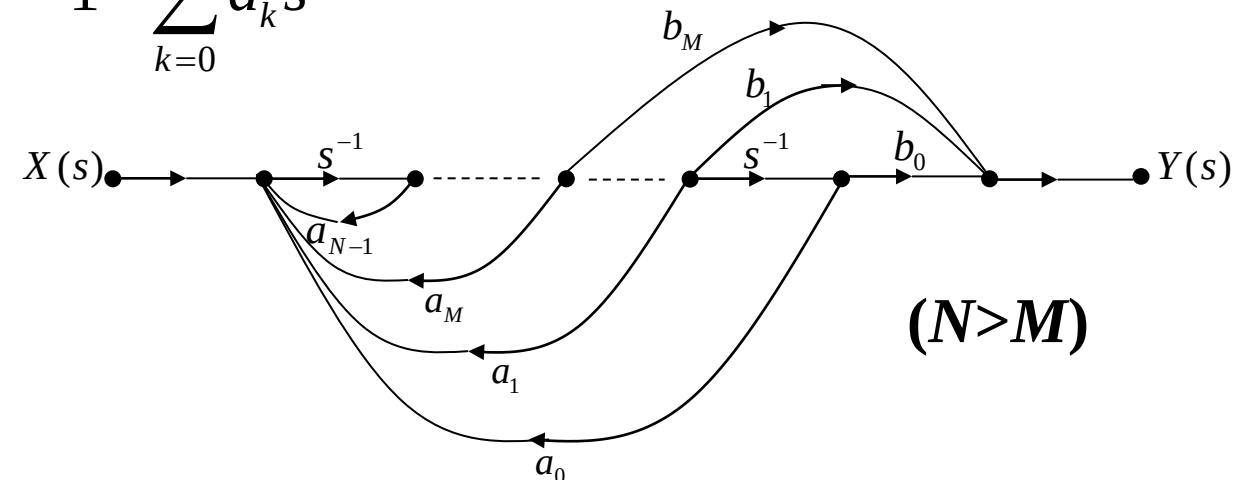
1. Phase Variable Method

The differential equation and system function of the system are as follows:

$$\frac{d^N y(t)}{dt^N} - \sum_{k=0}^{N-1} a_k \frac{d^k y(t)}{dt^k} = \sum_{r=0}^M b_r \frac{d^r x(t)}{dt^r}$$

$$H(s) = \frac{\sum_{r=0}^M b_r s^r}{s^N - \sum_{k=0}^{N-1} a_k s^k} = \frac{\sum_{r=0}^M b_r s^{r-N}}{1 - \sum_{k=0}^{N-1} a_k s^{k-N}}$$

Signal flow graph:



8.2 Formulation of State Equations for Continuous-Time Systems

- (1) In the signal flow graph, **select the output of the integrator (from the output to the input direction) as the state variable;**
- (2) Formulate the state equations:

$$\frac{d\lambda_1(t)}{dt} = \lambda_2(t)$$

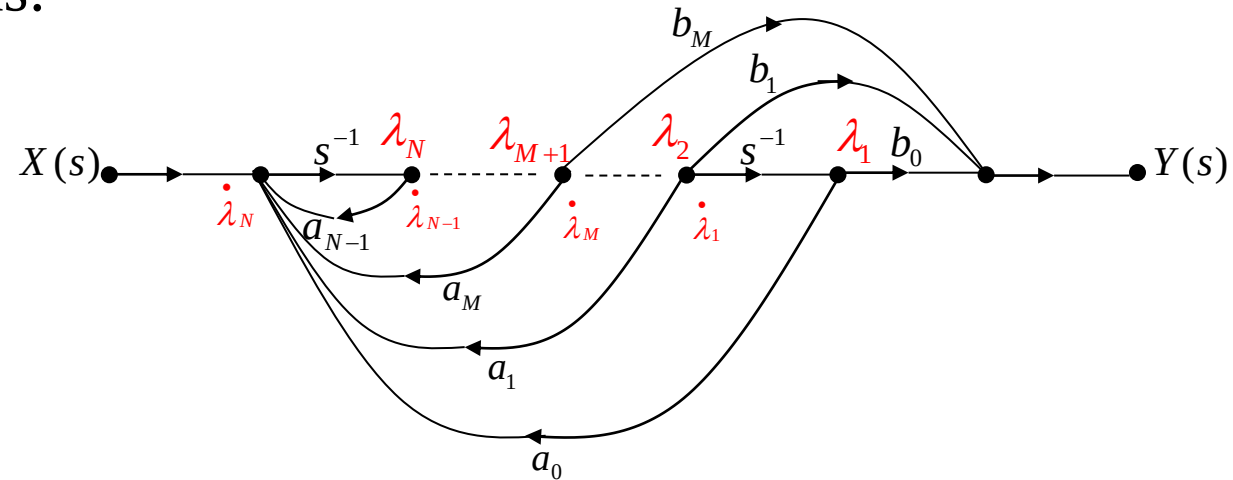
$$\frac{d\lambda_2(t)}{dt} = \lambda_3(t)$$

...

$$\frac{d\lambda_{k-1}(t)}{dt} = \lambda_k(t)$$

...

$$\frac{d\lambda_N(t)}{dt} = a_0\lambda_1(t) + a_1\lambda_2(t) + \dots + a_{N-1}\lambda_N(t) + x(t)$$



8.2 Formulation of State Equations for Continuous-Time Systems

Write in matrix form:

$$\underbrace{\begin{pmatrix} \frac{d\lambda_1(t)}{dt} \\ \frac{d\lambda_2(t)}{dt} \\ \vdots \\ \frac{d\lambda_{N-1}(t)}{dt} \\ \frac{d\lambda_N(t)}{dt} \end{pmatrix}}_{\lambda'(t)} = \underbrace{\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \cdots & & \cdots & & \vdots \\ 0 & \cdots & & \cdots & 0 & 1 \\ a_0 & a_1 & \cdots & \cdots & a_{N-2} & a_{N-1} \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \\ \vdots \\ \lambda_{N-1}(t) \\ \lambda_N(t) \end{pmatrix}}_{\lambda(t)} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}}_{\mathbf{B}} [x(t)]$$

State equation: $\lambda'(t) = \mathbf{A}\lambda(t) + \mathbf{B}x(t)$

The N -th row of matrix $\mathbf{A}$ is formed by multiplying the coefficients on the left side of the input-output equation by -1 and arranging them in reverse order; the elements immediately above the main diagonal are all 1, and the rest are 0.

Matrix $\mathbf{B}$ is a column matrix, where **the last element is 1** and the rest are 0.

8.2 Formulation of State Equations for Continuous-Time Systems

Output Equation: $\mathbf{y}(t) = \mathbf{C}\boldsymbol{\lambda}(t) + \mathbf{D}\mathbf{x}(t)$

According to the signal flow graph, **the output equation** (for the general case $N > M$) is:

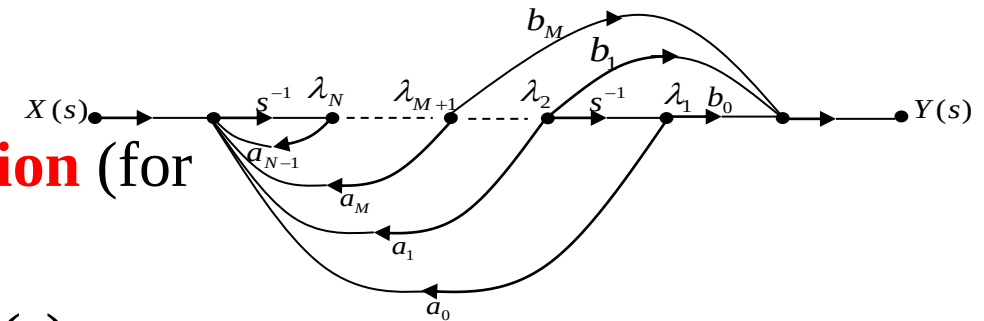
$$y(t) = b_0 \lambda_1(t) + b_1 \lambda_2(t) + \dots + b_{M-1} \lambda_M(t) + b_M \lambda_{M+1}(t)$$

Write it in matrix form:

$$y(t) = (b_0 \quad b_1 \quad \dots \quad b_{M-1} \quad b_M \quad 0 \quad \dots \quad 0) \begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \\ \dots \\ \lambda_M(t) \\ \lambda_{M+1}(t) \\ \dots \\ \lambda_N(t) \end{pmatrix}$$

$$\mathbf{C} = [b_0 \quad b_1 \quad \dots \quad b_{M-1} \quad b_M \quad 0 \quad \dots \quad 0]$$

$$\text{Matrix } \mathbf{D} = \mathbf{0}$$



Matrix $\mathbf{C}$ is a row vector: **the first $M+1$ elements are formed by reversing the coefficients on the right side of the input-output equation**, and the remaining elements are 0.

8.2 Formulation of State Equations for Continuous-Time Systems

If $N=M$, the signal flow graph of the system is:

At this point, **the state equation of the system remains the same as before**, but the **C** and **D** matrices in the output equation change:

$$y(t) = b_0 \lambda_1(t) + b_1 \lambda_2(t) + \dots$$

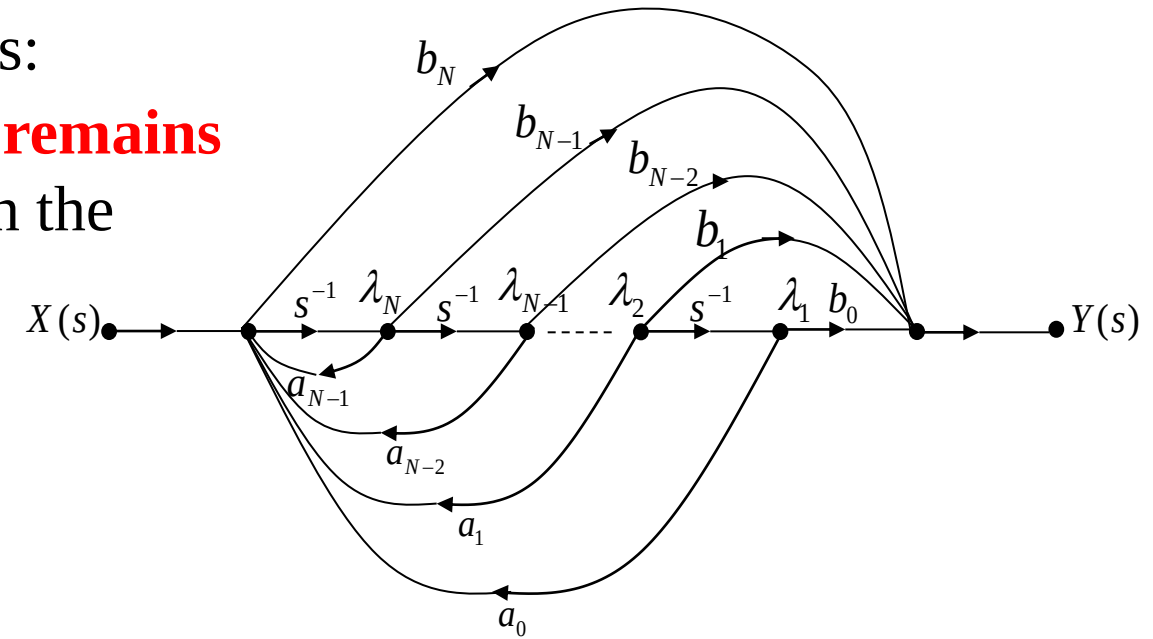
$$+ b_{N-1} \lambda_N(t) + b_N \frac{d\lambda_N(t)}{dt}$$

$$\therefore \frac{d\lambda_N(t)}{dt} = a_0 \lambda_1(t) + a_1 \lambda_2(t) + \dots + a_{N-1} \lambda_N(t) + x(t)$$

$$\therefore y(t) = (b_0 + b_N a_0) \lambda_1(t) + (b_1 + b_N a_1) \lambda_2(t) + \dots + (b_{N-1} + b_N a_{N-1}) \lambda_N(t) + b_N x(t)$$

Matrix **C** (at this point) $\mathbf{C} = [(b_0 + b_N a_0) \quad (b_1 + b_N a_1) \quad \dots \quad (b_k + b_N a_k) \quad \dots \quad (b_{N-1} + b_N a_{N-1})]$

Matrix **D** (at this point) $\mathbf{D} = b_N$

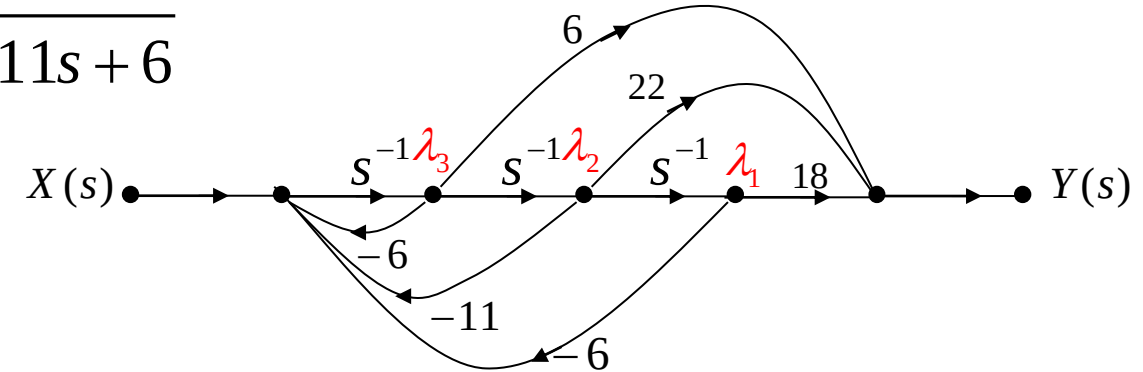


8.2 Formulation of State Equations for Continuous-Time Systems

Example 8-2: Given the system function below, use the phase variable method to formulate the system's state equations and output equations.

$$H(s) = \frac{6s^2 + 22s + 18}{s^3 + 6s^2 + 11s + 6}$$

Solution: $H(s) = \frac{6s^{-1} + 22s^{-2} + 18s^{-3}}{1 - (-6s^{-1} - 11s^{-2} - 6s^{-3})}$



Draw the system's signal flow graph. **Select the outputs of the integrators (in order from the output end to the input end) as the state variables.**

Formulate the state equations:

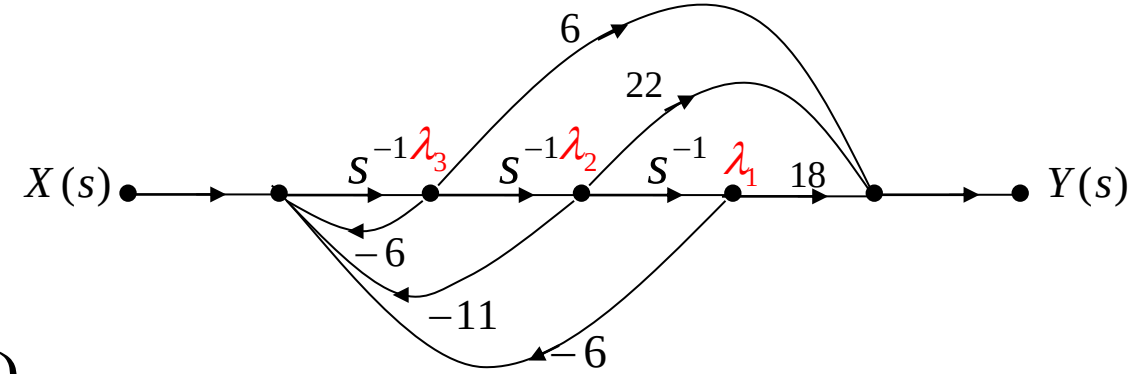
$$\frac{d\lambda_1(t)}{dt} = \lambda_2(t) \quad \frac{d\lambda_2(t)}{dt} = \lambda_3(t)$$

$$\frac{d\lambda_3(t)}{dt} = -6\lambda_1(t) - 11\lambda_2(t) - 6\lambda_3(t) + x(t)$$

8.2 Formulation of State Equations for Continuous-Time Systems

State equation in matrix form:

$$\begin{pmatrix} \frac{d\lambda_1(t)}{dt} \\ \frac{d\lambda_2(t)}{dt} \\ \frac{d\lambda_3(t)}{dt} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{pmatrix}}_A \begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \\ \lambda_3(t) \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_B x(t)$$



The system's output equation: $y(t) = 18\lambda_1(t) + 22\lambda_2(t) + 6\lambda_3(t)$

Output equation in matrix form: Matrices in the equation: $\mathbf{C} = (18 \quad 22 \quad 6) \quad \mathbf{D} = 0$

$$y(t) = (18 \quad 22 \quad 6) \begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \\ \lambda_3(t) \end{pmatrix}$$

Note: Based on the previous analysis, once the system equation or system function is known, the above matrices and equations can be written directly **without drawing the signal flow graph**.

Given the system function $H(s) = \frac{2s+6}{s^2+3s+2}$, the state equation matrix of the system is ()

- ☐ A $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- ☒ B $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- ☐ C $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 3 & 2 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- ☐ D $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$

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8.2 Formulation of State Equations for Continuous-Time Systems



Example 8-3: Given the system functions below, use the phase variable method to formulate the system's state equations and output equations.

$$H_1(s) = \frac{2s + 6}{s^2 + 3s + 2} \qquad H_2(s) = \frac{s^2 + 2s + 6}{s^2 + 3s + 2}$$

Solution: Based on the **phase variable method**, the matrices of the equations can be written directly.

$$1) \quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \mathbf{C} = (6 \quad 2) \quad \mathbf{D} = 0$$

2) **A and B are same as above.**

$$\mathbf{C} = (6 - 2 \quad 2 - 3) = (4 \quad -1) \quad \mathbf{D} = 1$$

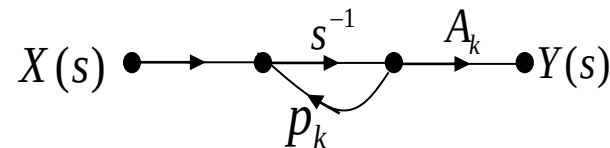
8.2 Formulation of State Equations for Continuous-Time Systems

2. Diagonal Method

Decompose the system function into partial fractions (expressed as the sum of first-order fractions):

$$H(s) = \frac{N(s)}{D(s)} = \sum_{k=1}^N \frac{A_k}{s - p_k}$$

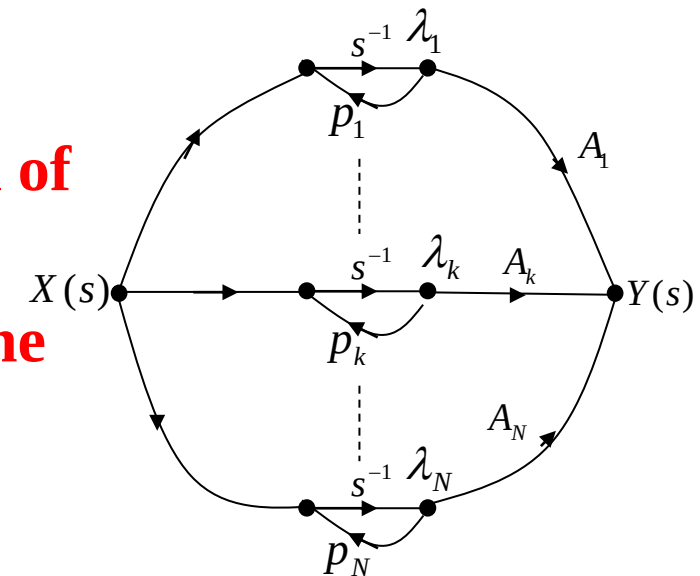
Each partial fraction corresponds to a first-order system. The signal flow graph of a first-order system is as follows:



The signal flow graph of the system is **the parallel connection of these first-order systems.**

Select the output of each integrator as the state variable, then **the state equations** can be formulated:

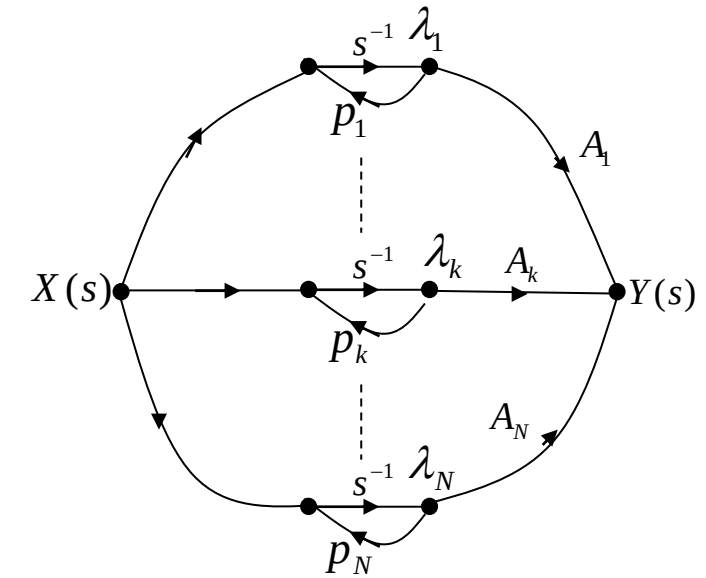
$$\frac{d\lambda_1(t)}{dt} = p_1\lambda_1(t) + x(t) \quad \dots \quad \frac{d\lambda_N(t)}{dt} = p_N\lambda_N(t) + x(t)$$



8.2 Formulation of State Equations for Continuous-Time Systems

Written in Matrix Form

$$\begin{pmatrix} \frac{d\lambda_1(t)}{dt} \\ \frac{d\lambda_2(t)}{dt} \\ \dots \\ \frac{d\lambda_{N-1}(t)}{dt} \\ \frac{d\lambda_N(t)}{dt} \end{pmatrix} = \begin{pmatrix} p_1 & & & 0 \\ & \dots & & \\ & & p_k & \\ & 0 & & \dots \\ & & & & p_N \end{pmatrix} \begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \\ \dots \\ \lambda_{N-1}(t) \\ \lambda_N(t) \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ \dots \\ 1 \\ 1 \end{pmatrix} [x(t)]$$



$$H(s) = \frac{N(s)}{D(s)} = \sum_{k=1}^N \frac{A_k}{s - p_k}$$

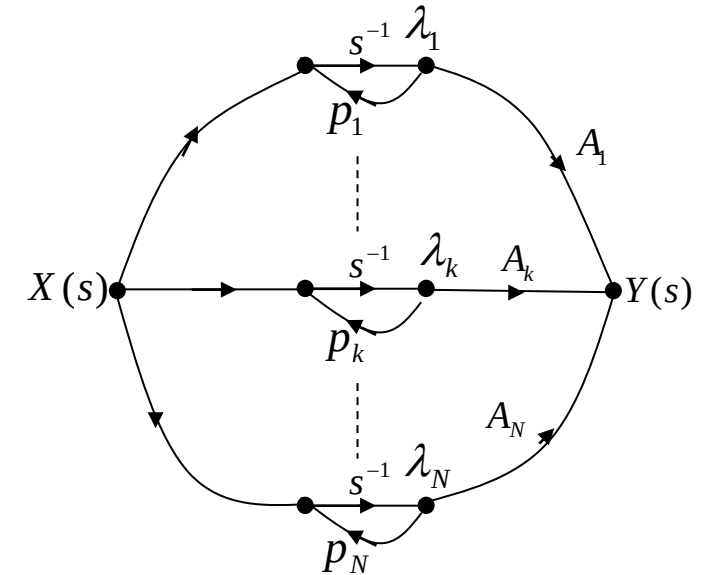
State equations with a **diagonal A-matrix** are of great significance in control theory research. In linear transformations, diagonalizing the **A-matrix** is a useful operation — this corresponds to the **parallel system** structure, where **state variables do not affect each other**, so the influence of system parameters on state variables can be studied independently.

8.2 Formulation of State Equations for Continuous-Time Systems

The **output equation of the system** is

$$y(t) = A_1 \lambda_1(t) + \dots + A_k \lambda_k(t) + \dots + A_N \lambda_N(t)$$

$$y(t) = \begin{pmatrix} A_1 & \dots & A_k & \dots & A_N \end{pmatrix} \begin{pmatrix} \lambda_1(t) \\ \dots \\ \lambda_k(t) \\ \dots \\ \lambda_N(t) \end{pmatrix}$$



8.2 Formulation of State Equations for Continuous-Time Systems



Example 8-4: Given the system function in Example 8-2, use the **Diagonal Method** to formulate the system's state equations and output equations.

$$H(s) = \frac{6s^2 + 22s + 18}{s^3 + 6s^2 + 11s + 6}$$

Solution: Decompose the system function into partial fractions,

$$H(s) = \frac{6s^2 + 22s + 18}{s^3 + 6s^2 + 11s + 6} = \frac{1}{s+1} + \frac{2}{s+2} + \frac{3}{s+3}$$

Thus, the coefficient matrices in the equations are:

$$\mathbf{A} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \mathbf{C} = (1 \quad 2 \quad 3) \quad \mathbf{D} = 0$$

State Equation: $\lambda'(t) = \mathbf{A}\lambda(t) + \mathbf{B}\mathbf{x}(t)$

Output Equation: $\mathbf{y}(t) = \mathbf{C}\lambda(t) + \mathbf{D}\mathbf{x}(t)$

8.3 Solution of State Equations for Continuous-Time Systems

8.3.1 Calculus and Laplace Transform of Vectors

Consider a time vector $\lambda(t)$, which is a column vector: $\lambda(t) = \begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \\ \dots \\ \lambda_N(t) \end{pmatrix}$

then

$$\frac{d}{dt}\lambda(t) = \begin{pmatrix} \frac{d\lambda_1(t)}{dt} \\ \frac{d\lambda_2(t)}{dt} \\ \dots \\ \frac{d\lambda_N(t)}{dt} \end{pmatrix} \quad \int_{-\infty}^t \lambda(\tau) d\tau = \begin{pmatrix} \int_{-\infty}^t \lambda_1(\tau) d\tau \\ \int_{-\infty}^t \lambda_2(\tau) d\tau \\ \dots \\ \int_{-\infty}^t \lambda_N(\tau) d\tau \end{pmatrix}$$

8.3 Solution of State Equations for Continuous-Time Systems



$$\Lambda(s) = \text{LT} \{ \lambda(t) \} = \int_{0^-}^{\infty} \lambda(t) e^{-st} dt = \begin{pmatrix} \int_{0^-}^{\infty} \lambda_1(t) e^{-st} dt \\ \int_{0^-}^{\infty} \lambda_2(t) e^{-st} dt \\ \dots \\ \int_{0^-}^{\infty} \lambda_N(t) e^{-st} dt \end{pmatrix} = \begin{pmatrix} \Lambda_1(s) \\ \Lambda_2(s) \\ \dots \\ \Lambda_N(s) \end{pmatrix}$$

$$\text{LT} \left\{ \frac{d}{dt} \lambda(t) \right\} = s\Lambda(s) - \lambda(0^-) = s \begin{pmatrix} \Lambda_1(s) \\ \Lambda_2(s) \\ \dots \\ \Lambda_N(s) \end{pmatrix} - \begin{pmatrix} \lambda_1(0^-) \\ \lambda_2(0^-) \\ \dots \\ \lambda_N(0^-) \end{pmatrix}$$

8.3 Solution of State Equations for Continuous-Time Systems

8.3.2 Laplace Transform Solution for State Equations

1. Solution of State Equations and Output Equations

$$\frac{d\lambda(t)}{dt} = \mathbf{A}\lambda(t) + \mathbf{B}\mathbf{x}(t)$$

Take the Laplace transform of both sides of the equation:

$$s\Lambda(s) - \lambda(0^-) = \mathbf{A}\Lambda(s) + \mathbf{B}\mathbf{X}(s)$$

$$(s\mathbf{I} - \mathbf{A})\Lambda(s) = \lambda(0^-) + \mathbf{B}\mathbf{X}(s)$$

Laplace transform of the state variable:

$$\Lambda(s) = (s\mathbf{I} - \mathbf{A})^{-1}\lambda(0^-) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{X}(s)$$

$$\lambda(t) = \mathcal{L}^{-1}\{\Lambda(s)\} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\lambda(0^-) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{X}(s)\}$$

$$= \underbrace{\mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\lambda(0^-)\}}_{\text{Zero-input Component}} + \underbrace{\mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\} * \mathbf{x}(t)}_{\text{Zero-state Component}}$$

$(s\mathbf{I} - \mathbf{A})^{-1}$ - **Characteristic Matrix of the System** $|s\mathbf{I} - \mathbf{A}|$ - **Characteristic Determinant of the System**

8.3 Solution of State Equations for Continuous-Time Systems

$$\mathbf{y}(t) = \mathbf{C}\boldsymbol{\lambda}(t) + \mathbf{D}\mathbf{x}(t)$$

Laplace transform of the output variable:

$$\mathbf{Y}(s) = \mathbf{C}\boldsymbol{\Lambda}(s) + \mathbf{D}\mathbf{X}(s)$$

$$\boldsymbol{\Lambda}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\lambda}(0^-) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{X}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\lambda}(0^-) + [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{X}(s)$$

then

$$\mathbf{y}(t) = \mathcal{L}^{-1}\{\mathbf{Y}(s)\} = \mathcal{L}^{-1}\{\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\lambda}(0^-) + [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{X}(s)\}$$

$$= \underbrace{\mathcal{L}^{-1}\{\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\lambda}(0^-)\}}_{\text{Zero-input Component}} + \underbrace{\mathcal{L}^{-1}\{\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}\}}_{\text{Zero-state Component}} * \mathbf{x}(t)$$

$(s\mathbf{I} - \mathbf{A})^{-1}$ --Characteristic Matrix of the System

$|s\mathbf{I} - \mathbf{A}|$ --Characteristic Determinant of the System

8.3 Solution of State Equations for Continuous-Time Systems

2. State Transition Matrix

From the previous analysis, the Laplace transform of the system's state variable is:

$$\Lambda(s) = (s\mathbf{I} - \mathbf{A})^{-1}[\lambda(0^-) + \mathbf{B}\mathbf{X}(s)]$$

$$\lambda(t) = \mathcal{L}^{-1}\{\Lambda(s)\} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\} * \mathcal{L}^{-1}\{\lambda(0^-) + \mathbf{B}\mathbf{X}(s)\}$$

**State Transition Matrix:
inverse Laplace transform of
the characteristic matrix**


$$\Phi(t) = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$$

The state transition matrix converts the system's initial state and the effect of the excitation into the system's state at any time $t > 0$.

8.3 Solution of State Equations for Continuous-Time Systems

3. Find the System (Transfer) Function

From the previous analysis, the Laplace transform of the system's output vector is:

$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\lambda}(0^-) + [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{X}(s)$$

The **zero-state component** is:

$$\mathbf{Y}_{zs}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{X}(s) = \mathbf{H}(s)\mathbf{X}(s)$$

$$\mathbf{H}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$



System (Transfer) Function

When different state variables are used for the same system, the physical nature of the system is not affected, and the **system transfer function remains unchanged**.

$$\hat{\mathbf{H}}(s) = \hat{\mathbf{C}}(s\mathbf{I} - \hat{\mathbf{A}})^{-1}\hat{\mathbf{B}} + \hat{\mathbf{D}} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

8.3 Solution of State Equations for Continuous-Time Systems

8.3.3 Time-Domain Solution for State Equations

From the previous Laplace transform solution, the solution to the state equation is:

$$\lambda(t) = \mathcal{L}^{-1} \left\{ (s\mathbf{I} - \mathbf{A})^{-1} \right\} \lambda(0^-) + \mathcal{L}^{-1} \left\{ (s\mathbf{I} - \mathbf{A})^{-1} \right\} \mathbf{B} * \mathbf{x}(t)$$

The state transition matrix in the formula is expressed as a matrix exponential:

$$\Phi(t) = \mathcal{L}^{-1} \left\{ (s\mathbf{I} - \mathbf{A})^{-1} \right\} = e^{\mathbf{A}t}$$

Thus, the solution to the state equation is:

$$\lambda(t) = e^{\mathbf{A}t} \lambda(0^-) + e^{\mathbf{A}t} \mathbf{B} * \mathbf{x}(t)$$

The solution to the state equation (or output equation) consists of the zero-input solution and zero-state solution added together. The variation law of both parts is related to the matrix $e^{\mathbf{A}t}$, so $e^{\mathbf{A}t}$ reflects the essence of the system's state change and is called the **state transition matrix**.

Time-Domain Solution for the Output Equation

$$\mathbf{y}(t) = \mathbf{C} \mathcal{L}^{-1} \left\{ (s\mathbf{I} - \mathbf{A})^{-1} \right\} \boldsymbol{\lambda}(0^-) + [\mathbf{C} \mathcal{L}^{-1} \left\{ (s\mathbf{I} - \mathbf{A})^{-1} \right\} \mathbf{B} + \mathbf{D} \delta(t)] * \mathbf{x}(t)$$

$$= \mathbf{C} e^{\mathbf{A}t} \boldsymbol{\lambda}(0^-) + [\mathbf{C} e^{\mathbf{A}t} \mathbf{B} + \mathbf{D} \delta(t)] * \mathbf{x}(t)$$

 Zero-input Solution  Zero-state Solution

8.4 Formulation of State Equations for Discrete-Time Systems

8.4.1 State Variables of Discrete Systems

Initial State: Let the initial time be $n_0=0$, the initial state of an N -th order system usually refers to $y(-1)$, $y(-2)$, $\cdots$, $y(-N)$.

State variables of an N -th order system are denoted as:

$$\lambda_1(n) \text{ , } \lambda_2(n) \text{ , } \cdots \text{ , } \lambda_N(n)$$

State Vector:

$$\lambda(n) = \begin{bmatrix} \lambda_1(n) \\ \lambda_2(n) \\ \vdots \\ \lambda_N(n) \end{bmatrix}$$

Given the difference equation of a linear time-invariant causal system below. Assuming the system's initial state is zero, what are the coefficient matrices of the state equation and output equation? ()

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n-1)$$

- ☐ A $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1/8 & -1/4 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{C} = [0 \quad 1], \mathbf{D} = 0$
- ☐ B $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1/8 & -1/4 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{C} = [1 \quad 0], \mathbf{D} = 1$
- ☐ C $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1/8 & 1/4 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{C} = [0 \quad -1], \mathbf{D} = 1$
- ☒ D $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1/8 & 1/4 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{C} = [0 \quad 1], \mathbf{D} = 0$

提交

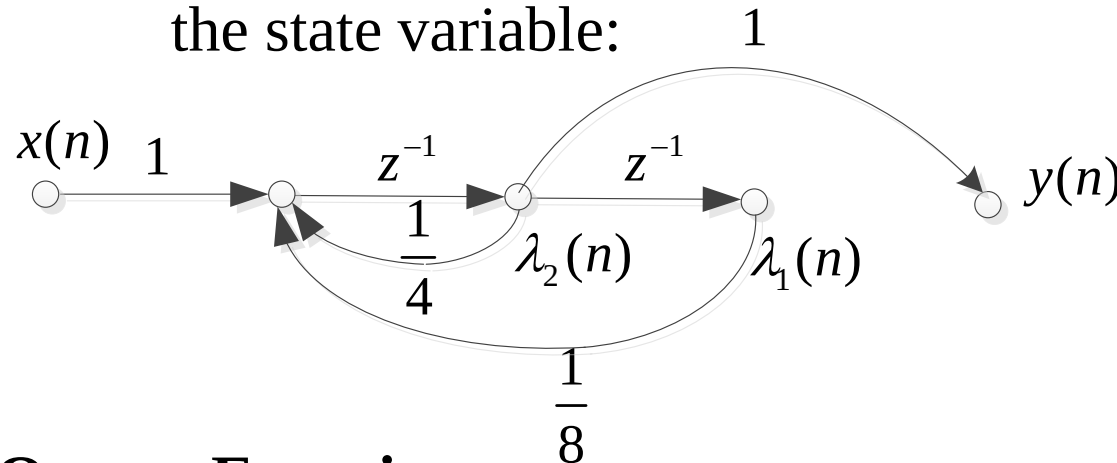
8.4 Formulation of State Equations for Discrete-Time Systems

8.4.2 State Equations and Output Equations of Discrete Systems

Example 8-5: Given the difference equation of a linear time-invariant causal system below, draw the signal flow graph, and formulate the state equation and output equation. Assume the system's initial state is zero.

$$y(n) - \frac{1}{4} y(n-1) - \frac{1}{8} y(n-2) = x(n-1)$$

Solution: Draw the signal flow graph, and select the output of the delay element as the state variable:



Output Equation:

$$y(n) = \lambda_2(n) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1(n) \\ \lambda_2(n) \end{bmatrix}$$

State Equation:

$$\lambda_1(n+1) = \lambda_2(n)$$

$$\lambda_2(n+1) = \frac{1}{4} \lambda_2(n) + \frac{1}{8} \lambda_1(n) + x(n)$$

$$\begin{bmatrix} \lambda_1(n+1) \\ \lambda_2(n+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{1}{8} & \frac{1}{4} \end{bmatrix} \begin{bmatrix} \lambda_1(n) \\ \lambda_2(n) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} x(n)$$

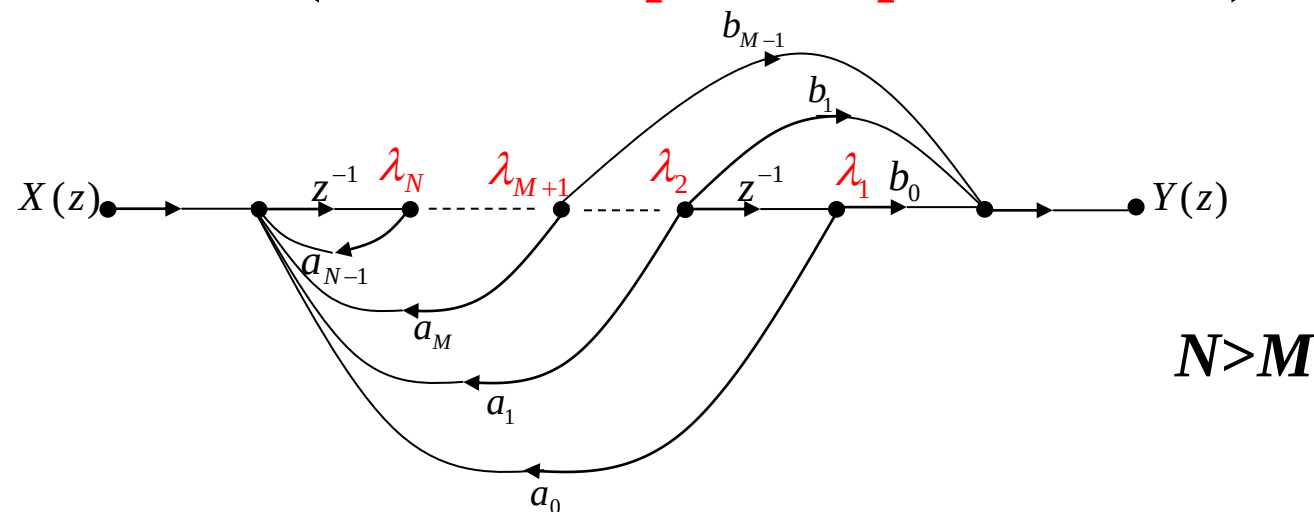
8.4 Formulation of State Equations for Discrete-Time Systems

Difference Equation & System Function of **Single-Input Single-Output** System:

$$y(n) - \sum_{k=0}^{N-1} a_k y(n - (N - k)) = \sum_{r=0}^M b_r x(n - (M - r))$$

$$H(z) = \frac{\sum_{r=0}^M b_r z^{-(M-r)}}{1 - \sum_{k=0}^{N-1} a_k z^{-(N-k)}}$$

(1) Draw the system's signal flow graph. In the graph, select **the output of the delay element** as the state variable (from the **output to input direction**)



8.4 Formulation of State Equations for Discrete-Time Systems

(2) Formulate the State Equation

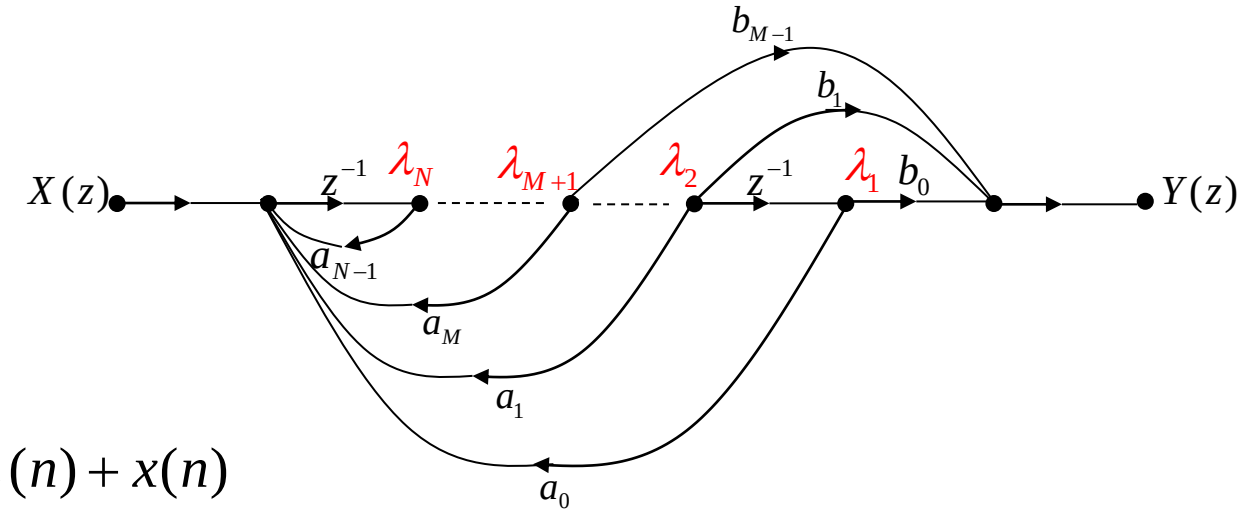
$$\lambda_1(n+1) = \lambda_2(n)$$

$$\lambda_2(n+1) = \lambda_3(n)$$

$\vdots$

$$\lambda_{N-1}(n+1) = \lambda_{N-2}(n)$$

$$\lambda_N(n+1) = a_0\lambda_1(n) + a_1\lambda_2(n) + \dots + a_{N-1}\lambda_N(n) + x(n)$$



Matrix form:

$$\underbrace{\begin{pmatrix} \lambda_1(n+1) \\ \lambda_2(n+1) \\ \vdots \\ \lambda_{N-1}(n+1) \\ \lambda_N(n+1) \end{pmatrix}}_{\lambda(n+1)} = \underbrace{\begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & \dots & \dots & \dots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & 0 & 1 \\ a_0 & a_1 & \dots & \dots & a_{N-2} & a_{N-1} \end{pmatrix}}_A \underbrace{\begin{pmatrix} \lambda_1(n) \\ \lambda_2(n) \\ \vdots \\ \lambda_{N-1}(n) \\ \lambda_N(n) \end{pmatrix}}_{\lambda(n)} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}}_B x(n)$$

State Equation: $\lambda(n+1) = A\lambda(n) + Bx(n)$

Matrix A and B are identical to the coefficient matrices in the state equation of continuous-time systems.

8.4 Formulation of State Equations for Discrete-Time Systems

Output Equation: $y(n) = \mathbf{C}\lambda(n) + \mathbf{D}x(n)$

If $N > M$,

$$y(n) = b_0\lambda_1(n) + b_1\lambda_2(n) + \dots + b_{M-1}\lambda_M(n) + b_M\lambda_{M+1}(n)$$

$$\mathbf{C} = [b_0 \quad b_1 \quad \dots \quad b_{M-1} \quad b_M \quad 0 \quad \dots \quad 0]$$

$$\mathbf{D} = 0$$

If $N = M$,

$$y(n) = (b_0 + b_N a_0)\lambda_1(n) + (b_1 + b_N a_1)\lambda_2(n) + \dots \\ + (b_{N-1} + b_N a_{N-1})\lambda_N(n) + b_N x(n)$$

$$\mathbf{C} = [(b_0 + b_N a_0) \quad (b_1 + b_N a_1) \quad \dots \quad (b_k + b_N a_k) \quad \dots \quad (b_{N-1} + b_N a_{N-1})]$$

$$\mathbf{D} = b_N$$

Matrix C and D are identical to the coefficient matrices in the output equation of continuous-time systems.

8.5 Solution of State Equations for Discrete-Time Systems

8.5.1 Time-Domain Solution of State & Output Equations

1. Solution of the State Equation

Assume an N -th order system with p inputs.

State equation: $\lambda(n+1) = \mathbf{A}\lambda(n) + \mathbf{B}\mathbf{x}(n)$

Substitute $\lambda(0)$ into the equation and iterate to obtain:

$$\begin{aligned}\lambda(n) &= \mathbf{A}^n \lambda(0) + \mathbf{A}^{n-1} \mathbf{B} \mathbf{x}(0) + \mathbf{A}^{n-2} \mathbf{B} \mathbf{x}(1) + \cdots + \mathbf{A} \mathbf{B} \mathbf{x}(n-2) + \mathbf{B} \mathbf{x}(n-1) \\ &= \mathbf{A}^n \lambda(0) + \sum_{i=0}^{n-1} \mathbf{A}^{n-1-i} \mathbf{B} \mathbf{x}(i) \\ &= \mathbf{A}^n \lambda(0) + (\mathbf{A}^{n-1} \mathbf{B}) * \mathbf{x}(n)\end{aligned}$$

Let $\Phi(n) = \mathbf{A}^n$  **State Transition Matrix: describes the state transition from the initial time from 0 to n.**

$$\lambda(n) = \underbrace{\Phi(n) \lambda(0)}_{\lambda_{zi}(n)} + \underbrace{\Phi(n-1) \mathbf{B} * \mathbf{x}(n)}_{\lambda_{zs}(n)}$$

(Zero-input Component) (Zero-state Component)

8.5 Solution of State Equations for Discrete-Time Systems

2. Solution of the Output Equation

Assume an N -th order system with p inputs and q outputs.

Output equation: $\mathbf{y}(n) = \mathbf{C}\lambda(n) + \mathbf{D}\mathbf{x}(n)$

Substitute $\lambda(n)$ into the output equation: $\mathbf{y}(n) = \mathbf{C}\Phi(n)\lambda(0) + \mathbf{C}\Phi(n-1)\mathbf{B} * \mathbf{x}(n) + \mathbf{D}\mathbf{x}(n)$

Introduce $\delta(n)$: $\delta(n) = \begin{bmatrix} \delta(n) & & 0 \\ & \delta(n) & \\ & & \ddots \\ 0 & & & \delta(n) \end{bmatrix}_{p \times p}$ then $\mathbf{D}\mathbf{x}(n) = \mathbf{D}\delta(n) * \mathbf{x}(n)$

$$\mathbf{y}(n) = \mathbf{C}\Phi(n)\lambda(0) + [\mathbf{C}\Phi(n-1)\mathbf{B} + \mathbf{D}\delta(n)] * \mathbf{x}(n) = \underbrace{\mathbf{C}\Phi(n)\lambda(0)}_{\mathbf{y}_{zi}(n)} + \underbrace{\mathbf{h}(n) * \mathbf{x}(n)}_{\mathbf{y}_{zs}(n)}$$

$\mathbf{h}(n) = \mathbf{C}\Phi(n-1)\mathbf{B} + \mathbf{D}\delta(n)$ — **Unit Sample Response Matrix** ($q \times p$)

$$= \begin{bmatrix} h_{11}(n) & h_{12}(n) & \cdots & h_{1p}(n) \\ \cdots & \cdots & & \cdots \\ h_{q1}(n) & h_{q2}(n) & \cdots & h_{qp}(n) \end{bmatrix}_{q \times p}$$

$h_{ij}(n)$: The unit impulse response of the output $y_i(n)$ when the input $x_j(n)$ acts alone.

8.5 Solution of State Equations for Discrete-Time Systems

8.5.2 z-Domain Solution of State Equations and Output Equations

1. z-Domain Solution of the State Equation

Let the initial time $n_0=0$

State equation: $\lambda(n+1) = \mathbf{A}\lambda(n) + \mathbf{B}\mathbf{x}(n)$

Left-shift property of the unilateral z-transform: $x(n+m) \longleftrightarrow z^m X(z) - \sum_{n=0}^m x(n)z^{m-n}$

Using the left-shift property, take the z-transform of both sides of the state equation:

$$z\Lambda(z) - z\lambda(0) = \mathbf{A}\Lambda(z) + \mathbf{B}\mathbf{X}(z)$$

$$\Lambda(z) = (z\mathbf{I} - \mathbf{A})^{-1} z\lambda(0) + (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B}\mathbf{X}(z)$$

$$\Phi(z) = (z\mathbf{I} - \mathbf{A})^{-1} z = (\mathbf{I} - z^{-1}\mathbf{A})^{-1}$$

← **Characteristic Matrix of the System** (z-transform of the state transition matrix)

$$\text{then } \Lambda(z) = \underbrace{\Phi(z)\lambda(0)}_{\Lambda_{zi}(z)} + \underbrace{z^{-1}\Phi(z)\mathbf{B}\mathbf{X}(z)}_{\Lambda_{zs}(z)}$$

$$\lambda(n) = ZT^{-1}[\Lambda(z)] = ZT^{-1}[\Lambda_{zi}(z)] + ZT^{-1}[\Lambda_{zs}(z)] = \lambda_{zi}(n) + \lambda_{zs}(n)$$

8.5 Solution of State Equations for Discrete-Time Systems

2. z-Domain Solution of the Output Equation

Let the initial time $n_0=0$

Output equation : $\mathbf{Y}(n) = \mathbf{C}\mathbf{x}(n) + \mathbf{D}\mathbf{x}(n)$

Take the unilateral z-transform of both sides:

$$\mathbf{Y}(z) = \mathbf{C}\mathbf{\Lambda}(z) + \mathbf{D}\mathbf{X}(z)$$

Substitute $\mathbf{\Lambda}(z)$ into the equation:

$$\begin{aligned}\mathbf{Y}(z) &= \mathbf{C}\mathbf{\Phi}(z)\boldsymbol{\lambda}(0) + [\mathbf{C}z^{-1}\mathbf{\Phi}(z)\mathbf{B} + \mathbf{D}]\mathbf{X}(z) \\ &= \underbrace{\mathbf{C}\mathbf{\Phi}(z)\boldsymbol{\lambda}(0)}_{\mathbf{Y}_{zi}(z)} + \underbrace{[\mathbf{C}z^{-1}\mathbf{\Phi}(z)\mathbf{B} + \mathbf{D}]\mathbf{X}(z)}_{\mathbf{Y}_{zs}(z)}\end{aligned}$$

$$\mathbf{H}(z) = \mathbf{C}z^{-1}\mathbf{\Phi}(z)\mathbf{B} + \mathbf{D} = ZT[\mathbf{h}(n)] \quad \leftarrow \text{System Function Matrix}$$

$$\mathbf{y}(n) = ZT^{-1}[\mathbf{Y}(z)] = ZT^{-1}[\mathbf{Y}_{zi}(z)] + ZT^{-1}[\mathbf{Y}_{zs}(z)] = \mathbf{y}_{zi}(n) + \mathbf{y}_{zs}(n)$$

Classified by the mathematical description method of the system:

- Input-Output Analysis Method:** An n -th order differential (difference) equation, suitable for single-input single-output systems (**Chapters 2, 3, 4, 6 and 7**)
- State Variable Analysis Method:** A system of n first-order differential (difference) equations, suitable for multi-input multi-output systems (**Chapter 8**)

Classified by the solution method of the system's mathematical model:

- Time-Domain Analysis Method:** Solve the response directly in the time domain without any transformation (**Chapters 2, 6 and 8**)
- Transform-Domain Analysis Method:** Transform the time functions of signals and system models into functions in the corresponding transform domain (e.g., **Fourier transform (Chapters 3 and 5), Laplace transform (Chapter 4), z-transform (Chapter 7), state variable analysis (Chapter 8)**)

1. Repeat all the examples in the lecture notes.
2. 2.71(a) (b) in the textbook.